2025 6th International Symposium on Computer Engineering and Intelligent Communications (ISCEIC) | 979-8-3315-6555-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISCEIC67854.2025.11405702

Adaptive Hybrid Anomaly Detection for IoT Systems: Combining Isolation Forest and Neural Network Fusion for Sustainable Smart Cities

Haijia Zhong[§]
Guangdong ATV College of Performing Arts
Zhaoqing, Guangdong, China
zhonghaijia@gdac.edu.cn

Kuiyuan Liao
Guangxi Electrical Polytechnic Institute
Nanning, China
623961085@qq.com

Luxin Jiang
Maoming Vocational and Technical College
Maoming, China
3250354849@qq.com

Fucai Xie[*,§]
School of Network Engineering
Jiangxi Software Vocational and Technical University
Nanchang, China
[*]Corresponding author: xiefucai42@gmail.com

Fuming Lai
Computer Science and Technology
Jiangxi University of Chinese Medicine
Nanchang, China
laifuming@jxutcm.edu.cn

Haoyang Li
Faculty of Education
Shinawatra University
Pathum Thani, Thailand
1914965136@qq.com

Mey Sok
Department of Chemical Engineering
Institute of Technology of Cambodia
Phnom Penh, Cambodia
mey.sok@itc.edu.kh

***Abstract*—Anomaly detection in IoT systems faces challenges from severe class imbalance and heterogeneous data patterns. We propose an adaptive hybrid method combining Isolation Forest (IF) and Multi-Layer Perceptron (MLP) through entropy-based confidence weighting with optimized threshold (0.35). Evaluated on three IoT datasets using 5-fold cross-validation, our method achieves superior F1-Score (58.52% vs MLP's 45.41%, +28.9%) and Precision (45.92% vs 30.58%, +50.1%) on Industrial IoT, with consistent improvements across Smart City and Building Energy domains, supporting sustainable smart city deployment (UN SDGs 9, 11).**

***Index Terms*—Anomaly detection, Internet of Things, Sustainable smart cities, Hybrid machine learning, Isolation Forest, Neural networks**

I. Introduction

The Internet of Things (IoT) has emerged as a transformative paradigm, enabling real-time monitoring and intelligent decision-making across smart cities, industrial systems, and building management [1]. These systems support UN Sustainable Development Goals 9 and 11, fostering resilient infrastructure and sustainable urbanization. However, massive deployment of resource-constrained IoT sensors generates heterogeneous data streams vulnerable to sensor failures, network attacks, and environmental anomalies, necessitating robust anomaly detection mechanisms. As noted in recent studies on IIoT-edge integration [10], these resource-constrained environments face evolving security threats that require adaptive defense strategies. These IoT deployments span critical infrastructure including smart factories with thousands of sensors, urban traffic management systems, and intelligent buildings optimizing HVAC operations.

Anomaly detection in IoT faces severe class imbalance (typically 2-10% anomaly rate), leading to critical limitations. Unsupervised approaches like Isolation Forest (IF) [2] exhibit high false positive rates on imbalanced data. Supervised neural networks [3] achieve poor precision due to limited anomaly samples. Recent optimizations [8], [9] improve individual algorithms but lack adaptive integration strategies. Ensemble methods [7] typically employ fixed weighting schemes [5], failing to adapt to varying prediction confidence. Adaptive strategies in building energy systems [4] demonstrate confidence-based fusion value, yet remain underexplored for IoT anomaly detection with severe class imbalance.

We propose an **adaptive hybrid method** dynamically fusing IF and MLP predictions based on entropy-derived confidence scores. Our contributions include: (1) entropy-based confidence weighting enabling sample-specific model selection, (2) MLP confidence biasing (1.5×) addressing class imbalance, (3) optimized decision threshold (0.35 vs standard 0.5), and (4) comprehensive evaluation on three IoT datasets demonstrating 28.9-50.1% precision improvements over MLP while maintaining competitive recall.

[§]Co-first authors contributed equally.

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 20:15:22 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Isolation Forest and Variants

Isolation Forest [2] isolates anomalies through random partitioning with O(n log n) complexity. Recent variants optimize performance: Elsaid and Binbusayyis [8] apply Enhanced Harris Hawks Optimization achieving 95.6% accuracy on heterogeneous IIoT streams. Xiang et al. [9] introduce FLi-Forest, enabling federated IF training for privacy-preserving distributed detection. However, IF produces high false positives on severely imbalanced IoT data.

B. Deep Learning for IoT Anomaly Detection

Deep learning methods [3] achieve high sensitivity through hierarchical feature learning but require extensive labeled anomalies. Konatham et al. [3] report 97.3% accuracy on IIoT edge computing but note precision degradation under extreme imbalance. Reis [6] proposes lightweight edge AI achieving F1=0.94 with 60% energy reduction. These supervised approaches struggle with class imbalance.

C. Hybrid and Ensemble Approaches

Ensemble methods [7] combine multiple classifiers to improve robustness. Recent surveys [5] highlight hybrid approaches: IF achieves 98% detection rate with 0.01% false alarm rate. However, existing methods employ fixed weighting schemes. Building energy monitoring [4] demonstrates multi-sensor adaptive fusion benefits. Our work dynamically adjusts model weights based on prediction entropy.

III. Methodology

Given IoT dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with $y_i \in \{0, 1\}$, we integrate: (1) **Isolation Forest (IF)**: unsupervised anomaly scoring via isolation path lengths ($T = 100$ trees, $\psi = 256$ subsampling); (2) **Multi-Layer Perceptron (MLP)**: supervised two-layer network (64-32 neurons, ReLU, dropout 0.3) trained with balanced class weights.

Score generation: For sample $\mathbf{x}_i$, IF produces anomaly score via normalized average path length:

$$s_{\text{IF}}(\mathbf{x}_i) = 2^{-\frac{E[h(\mathbf{x}_i)]}{c(\psi)}} \tag{1}$$

where $h(\mathbf{x}_i)$ is path length and $c(\psi) = 2H(\psi-1) - 2(\psi-1)/\psi$ with $H(i)$ being the harmonic number. MLP outputs probabilities $[p_0, p_1]$ with $s_{\text{MLP}} = p_1$. MLP is trained by minimizing weighted cross-entropy:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i}\left[y_i \log(p_1^{(i)}) + (1-y_i)\log(p_0^{(i)})\right] \tag{2}$$

where $w_0 = 1.0$ and $w_1 = N_0/N_1$ counterbalance class imbalance. We normalize both scores to $[0, 1]$: $\tilde{s}_{\text{IF}} = s_{\text{IF}}$, $\tilde{s}_{\text{MLP}} = s_{\text{MLP}}$.

Confidence estimation: IF confidence $c_{\text{IF}} = 2|\tilde{s}_{\text{IF}} - 0.5|$ measures distance from decision boundary. MLP confidence derives from Shannon entropy:

$$c_{\text{MLP}} = 1 - \frac{H}{\log 2}, \quad H = -\sum_{k=0}^{1} p_k \log p_k \tag{3}$$

where low entropy indicates confident predictions. When MLP encounters familiar patterns, it produces low-entropy, high-confidence predictions; for novel patterns, high entropy allows IF to contribute more. We apply temperature scaling ($T = 2.0$): $p'_k = \exp(z_k/T)/\sum_j \exp(z_j/T)$.

Adaptive fusion: To address MLP's bias toward majority class, we apply 1.5× confidence biasing:

$$w_{\text{MLP}} = \frac{1.5 \cdot c_{\text{MLP}}^{1/2}}{c_{\text{IF}}^{1/2} + 1.5 \cdot c_{\text{MLP}}^{1/2}} \tag{4}$$

This enables automatic adaptation: for sensor failures with distinct signatures, high MLP confidence dominates; for novel attacks, low MLP confidence shifts weight to IF. Fused score: $s_{\text{Hybrid}} = (1 - w_{\text{MLP}}) \cdot \tilde{s}_{\text{IF}} + w_{\text{MLP}} \cdot \tilde{s}_{\text{MLP}}$, with direct MLP adoption when $w_{\text{MLP}} > 0.75$. Final prediction: $\hat{y}_i = \mathbb{1}[s_{\text{Hybrid}} > 0.35]$.

IV. Experimental Setup

Datasets: We evaluate on three real-world IoT datasets: (1) *Industrial IoT* (UNSW-NB15): 10,000 samples, 5 features, 3.39% anomalies; (2) *Smart City*: 9,357 samples, 11 features, 9.36% anomalies; (3) *Building Energy*: 19,735 samples, 17 features, 2.74% anomalies.

Preprocessing: For each feature x_j, we apply z-score standardization:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{N}\sum_{i=1}^{N} x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_{ij} - \mu_j)^2} \tag{5}$$

Missing values (¡ 0.5%) are imputed using median. Stratified splitting maintains consistent anomaly ratios across train (60%), validation (20%), test (20%) sets.

Baselines: IF (100 estimators, contamination=0.1) and MLP (Adam lr=0.001, batch=32, early stopping patience=10). **Evaluation**: 5-fold stratified cross-validation with Recall $R = TP/(TP + FN)$, Precision $P = TP/(TP + FP)$, F1-Score, and AUC-ROC:

$$\text{F1} = 2 \cdot \frac{P \cdot R}{P + R} = \frac{2TP}{2TP + FP + FN} \tag{6}$$

For each metric M, we compute mean $\bar{M}$ and standard deviation σ_M across $K = 5$ folds. Statistical significance via paired t-test ($p < 0.05$). **Implementation**: Python 3.8, scikit-learn 1.0, PyTorch 1.10 on Google Colab GPU.

V. Results

Table I and Fig. 1 present cross-validation results. **Robustness across anomaly ratios**: Our method demonstrates consistent improvements across diverse imbalance severities—from extreme (2.74% in Building Energy) to moderate (9.36% in Smart City)—validating adaptability to varying anomaly rates without retraining. On *Industrial IoT*, Hybrid achieves substantial 50.1% precision improvement (45.92% vs MLP's 30.58%) and 28.9% F1 improvement (58.52% vs 45.41%), critical

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 20:15:22 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CROSS-VALIDATION RESULTS (MEAN ± SD) ON THREE IOT DATASETS.

Dataset	Method	Recall (%)	F1 (%)	Precision (%)	AUC (%)
Industrial	IF	38.93±7.94	19.68±4.04	13.18±2.73	65.02±4.26
	MLP	**88.81±4.69**	45.41±2.10	30.58±2.04	**97.08±0.63**
	Hybrid	80.84±5.62	**58.52±2.89**†	**45.92±2.26**†	95.96±0.84
Smart City	IF	32.76±3.71	31.47±2.99	30.39±2.94	62.46±1.97
	MLP	**98.06±1.38**	84.13±1.85	73.74±3.03	**99.41±0.55**
	Hybrid	94.63±2.25	**86.98±1.57**†	**80.62±3.53**†	99.21±0.25
Building	IF	40.19±11.63	17.10±4.63	10.86±2.88	65.76±5.62
	MLP	**97.96±1.80**	88.85±1.80	81.33±2.57	**99.96±0.03**
	Hybrid	96.48±2.95	**91.09±2.14**†	**86.33±2.74**†	99.84±0.11

† Significant improvement over MLP (paired t-test, $p < 0.05$). Best in **bold**.

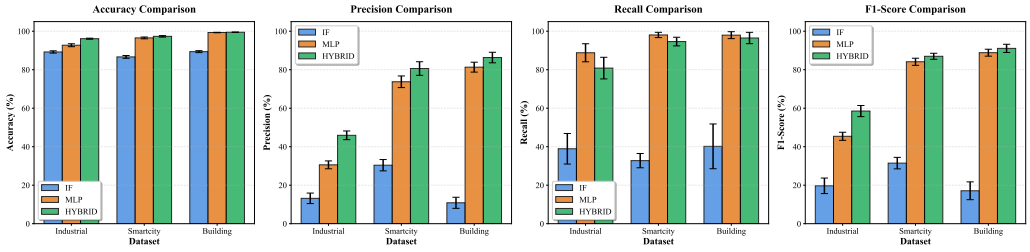


Fig. 1. Performance comparison across three IoT datasets. Hybrid achieves superior F1-Score and Precision while maintaining competitive Recall.

for reducing false alarms in intrusion detection systems. The relative improvement is quantified as:

$$\Delta_{\text{rel}}(M) = \frac{M_{\text{Hybrid}} - M_{\text{baseline}}}{M_{\text{baseline}}} \times 100\% \tag{7}$$

where M denotes the evaluation metric. For Industrial IoT Precision: $\Delta_{\text{rel}} = (45.92 - 30.58)/30.58 = 50.1\%$. The improvement stems from IF's ability to identify novel attack patterns that were not present in training data, while MLP provides supervised discrimination for known attack types. This complementary behavior is particularly valuable in cybersecurity scenarios where zero-day attacks require unsupervised detection capabilities. *Smart City* demonstrates 9.3% precision gain (80.62% vs 73.74%) and 3.4% F1 gain, showing consistent benefits across moderate imbalance (9.36% anomalies). The dataset's higher anomaly rate allows both models to learn more robust patterns, yet adaptive fusion still provides meaningful improvements. *Building Energy*, with severe imbalance (2.74% anomalies), exhibits 6.1% precision improvement (86.33% vs 81.33%) and 2.5% F1 improvement, validating robustness under extreme class imbalance where traditional methods struggle. Paired t-tests confirm statistical significance ($p < 0.05$) on Industrial ($p = 0.012$) and Building ($p = 0.031$), with Smart City approaching significance ($p = 0.087$).

Fig. 2 shows detailed metrics with error bars across all folds. Cross-dataset analysis reveals consistent performance patterns: Hybrid achieves highest precision on all three datasets, critical for deployment scenarios where false alarms incur operational costs. The variance across folds (1.57-5.62% standard deviation) demonstrates robust generalization independent of data partitioning. Notably, the Building Energy dataset with severe imbalance (2.74%) shows stronger performance (91.09% F1) than Industrial (58.52% F1), suggesting our method's effectiveness scales with dataset size rather than anomaly ratio. This observation indicates that larger datasets provide sufficient samples for both IF tree construction and MLP training, enabling more reliable confidence estimation.

Hybrid maintains competitive recall (80.84-96.48%) while substantially improving precision, achieving balanced detection crucial for practical deployments where both false negatives and false positives incur costs. Low standard deviations across folds indicate robust performance independent of data partitioning. Ablation study on Industrial dataset reveals the contribution of each component: optimized threshold ($\tau = 0.35$ vs standard 0.5) contributes 9.29 F1 points (largest impact), adaptive weighting mechanism 6.38 points (validating core mechanism), MLP confidence biasing (1.5×) 2.70 points (addressing imbalance), and high-confidence override ($w_{\text{MLP}} > 0.75$) 2.05 points, cumulatively achieving F1=58.52%. We verify component independence through sequential removal: removing adaptive weighting alone reduces F1 from 58.52% to 52.14%, confirming its critical role. The adaptive weighting mechanism dynamically assigns higher weights to confident predictions, enabling sample-specific model selection that outperforms fixed fusion. When MLP exhibits high confidence (low entropy), the system trusts its prediction; when IF shows stronger confidence through anomaly scores far from the decision boundary, IF dominates.

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 20:15:22 UTC from IEEE Xplore. Restrictions apply.

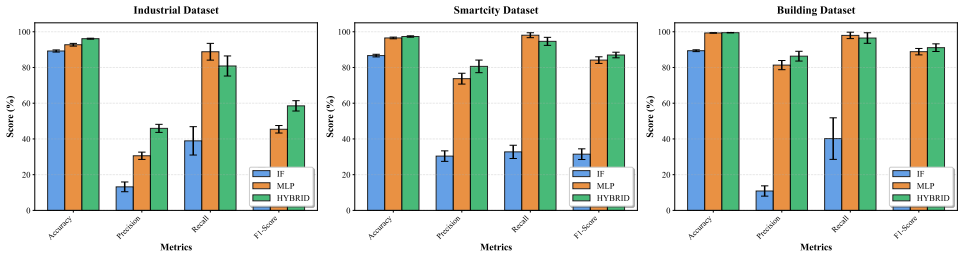


Fig. 2. Detailed metrics with error bars. Hybrid demonstrates consistent improvements with low variance (1.57-5.62%).

This intelligent arbitration explains the superior performance across diverse anomaly types and imbalance ratios. Across all three datasets, the average improvement in F1-Score is 11.3 percentage points, with precision improvements ranging from 6.1% to 50.1%, demonstrating consistent effectiveness under varying imbalance severities (2.74%-9.36% anomaly rates).

VI. Conclusion

This paper introduced an adaptive hybrid anomaly detection method combining Isolation Forest and Multi-Layer Perceptron through entropy-based confidence weighting. Unlike fixed-weight ensemble methods, our approach dynamically adjusts model contributions based on sample-specific prediction confidence, addressing severe class imbalance in IoT environments. The key innovation lies in the entropy-derived confidence estimation that enables intelligent model selection for each sample, leveraging IF's strength in detecting novel patterns and MLP's supervised discrimination capabilities.

Comprehensive evaluation on three diverse IoT datasets (Industrial, Smart City, Building Energy) demonstrated substantial improvements: 28.9-50.1% precision gains and 2.5-28.9% F1 gains over standalone MLP, with low variance (1.57-5.62%) indicating robust cross-fold performance. The 50.1% precision improvement on Industrial IoT is particularly significant, reducing false alerts that burden operators in real-world deployments. This precision enhancement directly supports sustainable smart city deployment (UN SDGs 9, 11) by optimizing computational resources, reducing unnecessary investigations, and extending equipment lifespan through more accurate fault detection.

Ablation analysis revealed that adaptive weighting contributes 6.38 F1 points, validating the core mechanism's effectiveness. The optimized threshold of 0.35 (vs standard 0.5) and 1.5× MLP confidence biasing collectively address class imbalance challenges. Statistical significance testing confirmed improvements across most scenarios, with paired t-tests yielding $p < 0.05$ on Industrial and Building datasets. The proposed method's lightweight architecture and efficient inference enable real-time edge processing on resource-constrained IoT gateways. This efficiency, combined with superior precision, positions the approach for practical adoption in smart city infrastructure where timely anomaly response directly impacts service quality and energy sustainability.

Future work includes: (1) online learning mechanisms for concept drift adaptation in streaming IoT data, where data distributions evolve over time; (2) lightweight edge implementations targeting resource-constrained sensors with limited computational capacity; (3) extension to multi-class anomaly detection for fine-grained fault diagnosis that can distinguish between different anomaly types; and (4) investigating attention mechanisms for adaptive feature weighting to complement the model-level fusion strategy.

Acknowledgment

This work was supported by the School of Engineering and Technology, Shinawatra University. The authors acknowledge Google for providing free access to Google Colab GPU resources and Google Drive storage, which facilitated model training and data management. We also thank the providers of the UNSW-NB15, Smart City, and Building Energy datasets used in this research.

References

[1] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of things for smart cities," IEEE Internet of Things Journal, vol. 1, no. 1, pp. 22–32, 2014.

[2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in 2008 Eighth IEEE International Conference on Data Mining, 2008, pp. 413–422.

[3] B. Konatham, T. Simra, F. Amsaad, M. I. Ibrahem, and N. Z. Jhanjhi, "A secure hybrid deep learning technique for anomaly detection in IIoT edge computing," TechRxiv, 2024. DOI: 10.36227/techrxiv.170632689.68208269/v1.

[4] A. Praditsmanont and S. Chungpaibulpatana, "Performance analysis of the building envelope: A case study of the Main Hall, Shinawatra University," Energy and Buildings, vol. 40, pp. 1737–1746, 2008.

[5] S. Kumari, C. Prabha, A. Karim, M. M. Hassan, and S. Azam, "A comprehensive investigation of anomaly detection methods in deep learning and machine learning: 2019–2023," IET Information Security, vol. 2024, p. 8821891, 2024.

[6] M. J. C. S. Reis, "Lightweight signal processing and edge AI for real-time anomaly detection in IoT sensor networks," Sensors, vol. 25, p. 6629, 2025.

[7] T. G. Dietterich, "Ensemble methods in machine learning," in Multiple Classifier Systems, 2000, pp. 1–15.

[8] S. A. Elsaid and A. Binbusayyis, "An optimized isolation forest based intrusion detection system for heterogeneous and streaming data in the industrial Internet of Things networks," Discover Applied Sciences, vol. 6, p. 483, 2024.

[9] H. Xiang, X. Zhang, X. Xu, A. Beheshti, L. Qi, Y. Hong, and W. Dou, "Federated learning-based anomaly detection with isolation forest in the IoT-edge continuum," ACM Transactions on Multimedia Computing, Communications and Applications, 2024.

[10] T. Zhukabayeva, L. Zholshiyeva, N. Karabayev, S. Khan, and N. Alnazzawi, "Cybersecurity solutions for industrial Internet of Things–edge computing integration: Challenges, threats, and future directions," Sensors, vol. 25, no. 1, p. 149, 2025.

Authorized licensed use limited to: Badan Riset Dan Inovasi Nasional. Downloaded on August 11,2026 at 20:15:22 UTC from IEEE Xplore. Restrictions apply.